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Hydrogen production performance of lithium hydride doped with solid acids or strong acid–weak base salts

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ABSTRACT

Lithium hydride (LiH) is a promising material for the production of hydrogen via hydrolysis. However, the formation of lithium hydroxide (LiOH) during the hydrolysis process hinders contact between the remaining LiH and water, retarding the hydrolysis reaction and increasing the actual amount of water required. Through this study, it is demonstrated that an increase in the hydrogen ion concentration can improve the hydrolysis of LiH. When the hydrogen ion concentration reaches 1 mol L^{-1}, the maximum hydrolysis rate is 232.6 mL min^{-1}, and the H_2 generation yield reaches 2102.8 mL. Additionally, we investigated the effects of doping solid acids and strong acid–weak base salts on LiH hydrolysis. Experimental results indicate that the addition of solid acids led to a hydrogen generation behavior similar to that with acid solutions. In particular, after the addition of oxalic acid, the maximum hydrolysis rate attained was 362.0 mL min^{-1}, and the H_2 generation yield reached 2108.3 mL. After the addition of DL-malic acid, citric acid and DL-tartaric acid, the maximum hydrolysis rate attained were 173.1, 170.0 and 255.8 mL min^{-1}, respectively. Under alkaline conditions, the hydrolysis of strong acid–weak base salts was enhanced, and the released hydrogen ions consumed LiOH, thereby increasing the rate of LiH hydrolysis. After the addition of $CuSO_4$, the maximum hydrolysis rate attained was 403.8 mL min^{-1}, and the H_2 generation yield attaining 2036.0 mL. After the addition of $AlCl_3$ and $FeCl_3$, the maximum hydrolysis rate attained were 212.5 and 221.3 mL min^{-1}. Thus, an increase in the concentration of hydrogen ions and the addition of solid acids and strong acid–weak base salts are effective techniques for reducing the adverse effects of LiOH formation on LiH hydrolysis for hydrogen production.

1. Introduction

The depletion of fossil fuels and the issue of global environmental pollution have become critical concerns that need to be immediately addressed. However, with progress in science and technology and population increase, the energy demand has increased significantly [1–3]. Replacing fossil fuels with green renewable energy sources has become a major research direction. Hydrogen energy, a renewable, high-calorific, and pollution-free energy, is considered a promising alternative to fossil energy [4,5]. Direct combustion of hydrogen in internal combustion engines and hydrogen fuel cells are the two primary energy conversion applications of hydrogen energy. Unlike internal combustion engines, hydrogen fuel cells can directly convert chemical energy into electrical energy without intermediate conversion [6].

Various types of hydrogen fuel cells include proton exchange membrane fuel cells (PEMFCs), phosphoric acid fuel cells, alkaline fuel cells, molten carbonate fuel cells, and solid oxide fuel cells. Among them, PEMFCs are widely used in transportation [7], off-grid power supplies, portable power supplies, and other fields, owing to their low-temperature operation, high power density, and fast start-up [8]. Depending solely on the amount of feed hydrogen, fuel cells can provide a longer power supply for portable electronic products such as unmanned aerial vehicles (UAVs) and laptops compared to conventional lithium-ion batteries. However, the most common method of hydrogen storage is a high-pressure hydrogen tank, which require auxiliary devices for compression and gas control. This reduces the volume and mass efficiency of fuel cells, limiting the development of portable energy. Under such conditions, on-demand and real-time hydrogen production appears to be a potential alternative to high-pressure hydrogen storage. Devices based on hydrogen storage alloys and methods such as the reformation of methanol and hydrolysis of hydrogen storage materials have been used in portable fuel-cell systems [9–12]. The hydrolysis of

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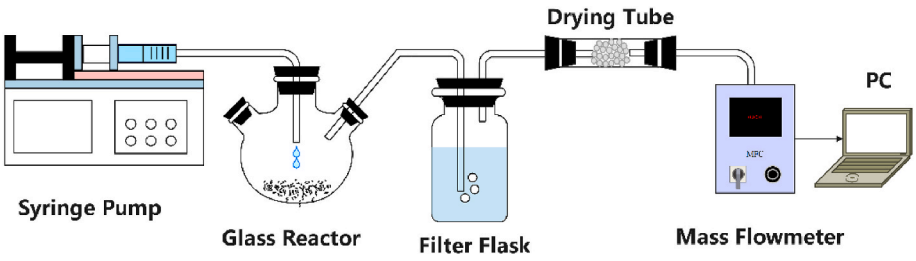


Fig. 1. Schematic of equipment used for evaluating hydrolytic performance.

hydrogen storage materials can be performed under moderate conditions and has thus attracted attention [3,13–15].

Among the hydrogen storage materials, lithium hydride (LiH) has several advantages as a hydrogen source. For example, LiH has a large theoretical hydrogen capacity (12 wt%) [16], and its relative fraction mass is only 7.949. Moreover, LiH can rapidly react with water without requiring a catalyst or intricate conditions. However, because of the rapid reaction between liquid water and LiH, the hydrogen production rate cannot easily be controlled.

Studies have found that hydrogen production is easier to control when water vapor is used instead of liquid water [17]. To regulate the rate of the hydrolysis reaction, several studies have been conducted on the hydrolysis mechanism and kinetics of LiH steam hydrolysis, reporting that lithium compounds such as lithium hydroxide (LiOH), lithium oxide (Li_2O), and lithium hydroxide monohydrate ($LiOH{\cdot}H_2O$) are obtained as hydrolysis products [18]. These products have been characterized by several detection techniques including scanning electron microscopy (SEM), atomic force microscopy (AFM) [19–22], Rutherford backscattering spectroscopy (RBS) [20], diffuse reflectance infrared Fourier transform (DRIFT) [23,24], X-ray diffraction (XRD) powder analysis [25–29], Raman spectroscopy [22,27,30,31], infrared spectroscopy (FT-IR) [28,29], X-ray photoelectron spectroscopy (XPS) [29,32,33], neutron radiography, and neutron computed tomography [34]. Results have shown that the formation of hydrolysis products depends on the experimental conditions, especially the partial pressure of water vapor. However, the order in which oxides and hydroxides appear is yet to be determined.

The effects of temperature, water vapor pressure, the amount of LiH material, and LiH powder packing conditions on the reaction rate and built hydrogen generators have also been explored [35–38]. Gislon et al. [26] designed a hydrogen generator via LiH steam hydrolysis. The hydrogen production rate of the reactor was 90%, and 50 g of LiH was hydrolyzed in 4 h. Strawser et al. [39] designed a passively controlled hydrogen generator using water vapor from the air as the water source for the hydrolysis of LiH. The pressure changes within the generator controlled the flow of water vapor in the air through the proton exchange membrane, realizing passive control. Martin et al. [40] investigated the hydrolysis of LiH for hydrogen production by setting the temperature of the reaction at 150, 300, and 500 °C. Finally, a scheme was proposed for a UAV propulsion system.

In addition, studies have used LiH as an additive to improve the hydrolytic performance of other hydrolytic hydrogen production materials. Wu et al. [41] added 3% LiH to MgH_2. Their experimental results showed that this proportion of LiH could increase hydrogen production by 72%. Moreover, the addition of $MgCl_2$ into a hydrolysis solution not only enhanced the hydrolysis rate but also concluded the reaction process. Liu et al. [42] reported that ball milling aluminum powder mixed with various hydrides can effectively improve hydrolysis, and LiH was shown to have the best effect. The Al-30 mol% LiH milled for 3 h at 75 °C reached a hydrogen yield of 96.3%, equivalent to 1442.4 mL H_2 per gram of composite within 1 h. Liu et al. [43] investigated hydrogen generation in ball-milled Al–LiH–salt mixtures. The experimental result showed that KCl was the most effective salt. The optimum amounts of KCl and LiH added were 10 mol%, and a milling time of 10 h was optimal. Li et al. [44] prepared a series of Al–LiH–metal oxide composites and hydrogen-producing materials using high-energy ball milling. The experimental results showed that the Al–LiH–Ga_2O_3 composite exhibited good hydrogen production performance at 298.15 K, a conversion efficiency of 98.6%, and a maximum hydrogen production rate of 3837 mL g^{-1} min^{-1}.

While there is a significant amount of research on the end products of LiH hydrolysis, relatively few studies have focused on the factors affecting hydrogen production through the hydrolysis process. These studies typically use water vapor (as the water source) to control the rate of LiH hydrolysis. However, the process of generating water vapor consumes excessive energy. Further, prior research has reported that the formation of LiOH reduces the hydrolysis rate of LiH and increases water consumption. In this study, we investigated a novel approach to mitigate the impact of LiOH formation on hydrolysis performance. Instead of relying on water vapor, a trace amount of liquid water was used as the water source to minimize parasitic power losses. Additionally, the effect of acid solutions with varying hydrogen ion concentrations on the hydrolysis performance of LiH was studied. Solid acids and strong acid–weak base salts were introduced into LiH to mimic the effects of acid solutions, thus promoting LiH hydrolysis. This study devised a method to reduce the influence of LiOH formation on hydrolysis performance, effectively enhancing the rate of LiH hydrolysis.

2. Experimental details

2.1. Reagents and preparation of the LiH mixture

LiH powder (97%), DL-malic acid (99.5%), citric acid (≥99.5%), oxalic acid (98%), DL-tartaric acid (AR), $AlCl_3$ (99%), $FeCl_3$ (99%), $CuSO_4$ (99%), and anhydrous methanol (99.9%) were purchased from Maclin (Shanghai, China). A dilute sulfuric acid solution was purchased from Shenzhen Xinake Trading Co., LTD (Shenzhen, China). 0.15 g solid acid or strong acid-weak base salt was mixed in a mortar with 0.85 g LiH powder. The mixture was stirred for 15 min to ensure homogenous mixing. To avoid the effects of moisture and carbon dioxide in the air, all materials were prepared in an Ar-filled glove box.

2.2. Hydrogen generation and characterization

Fig. 1 shows the setup used for evaluating hydrolysis performance. 1 g mixture of LiH or pure LiH powder was added into a 50-mL three-mouth flask in the glove box. Next, deionized water or an acid solution was injected through an injection pump at constant flow rates into the flask for each test. A gas washing bottle, designed to both cool the gas and remove any dissolved LiOH in the water mist, was attached to the back of the reactor. Then, a drying tube was used to remove additional water vapor. The H_2 generation rate was recorded by a mass flow meter (Z714AGXA, HORIBA, Japan).

The XRD patterns were obtained using a powder diffractometer (Ultima IV, Rigaku) at a rate of 10° min^{-1} in the scattering angle (2θ) range of 10°–80°. Fourier transform infrared (FT-IR) spectra were obtained using an FT-IR spectrometer (TENSOR II, Bruker) equipped with an attenuated total reflection module. XPS analysis was performed using

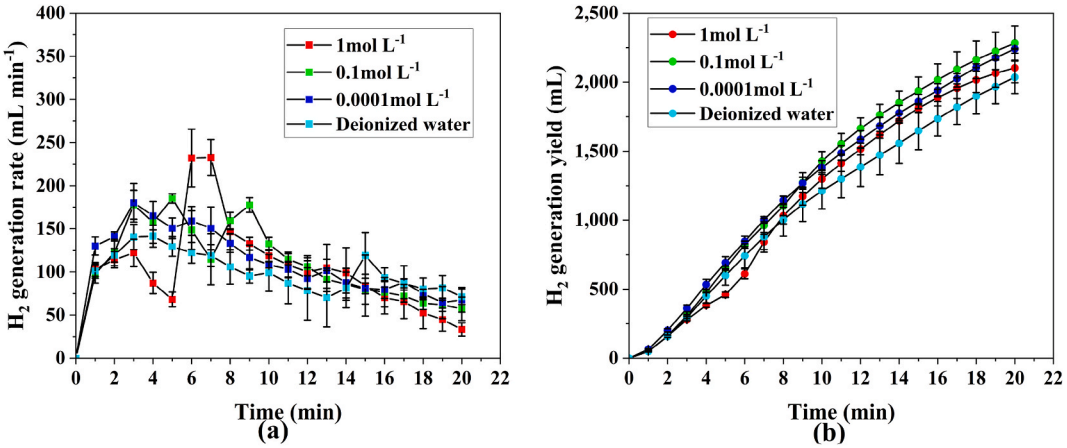


Fig. 2. (a) H_2 generation rate and (b) yield at different hydrogen ion concentrations as a function of time at 25 °C with a pump speed of 0.225 mL min^{-1}.

Table 1
Parameters of different solid acids.

Acid Name	Molecular Formula	Molecular Weight (g mol^{-1})	Density (g cm^{-3})	Type	Solubility at 20 °C (g/100 g water)
Oxalic Acid	$H_2C_2O_4$	90.03	1.900	Dibasic Acid	9.5
DL-Malic Acid	$C_4H_6O_5$	134.09	1.601	Dibasic Acid	55.8
Citric Acid	$C_6H_8O_7$	192.13	1.542	Tribasic Acid	59.0
DL-Tartaric Acid	$C_4H_6O_6$	150.09	1.697	Dibasic Acid	20.6

an ESCALAB 250XI (Thermo Scientific) spectrometer. Raman spectroscopy was performed using a confocal Raman microscope (LabRAM HR Evolution, HORIBA) using Ar^+ laser excitation with a wavelength of 532 nm.

3. Result and discussion

3.1. Hydrolysis at different hydrogen ion concentrations

The rate and yield of H_2 generation obtained at different hydrogen ion concentrations are shown in Fig. 2. The acidic solution consisted of a dilute sulfuric acid solution with hydrogen ion concentrations of c (H^+) = 1, 0.1, and 10^{-4} mol L^{-1}, with a liquid volume of 4.5 mL. The pumping rate of the peristaltic pump is 0.225 mL min^{-1}. Pure LiH was used as a reactant. Fig. 2a shows that as the concentration of hydrogen ions increased, the H_2 generation rate accelerated. However, with a further increase in the hydrogen ion concentration, a pattern of decrease following the increase in the H_2 generation rate emerged, with 1 mol L^{-1} hydrogen ions exhibiting the most notable behavior. At a concentration of 1 mol L^{-1}, the local hydrolysis rate is relatively fast, leading to the overgeneration of LiOH, which covers the surface of LiH and hinders the progression of the reaction. However, as the dropwise addition of the liquid continues, LiOH gradually dissolves, enabling the H_2 generation rate to rapidly increase, eventually to 232.6 mL min^{-1}.

Fig. 2b shows the variation in the H_2 generation yield with time. After the addition of the acidic solution, the total H_2 yield increased slightly. Under hydrogen ion concentrations of 1, 0.1, and 10^{-4} mol L^{-1}, the H_2 generation yields were 2102.8, 2282.7, and 2241.8 mL, respectively, which were higher than the yield of deionized water (2035.4 mL).

3.2. Hydrolysis for different solid acid types

As discussed in the previous section, different concentrations of acidic solution can affect the hydrolysis performance of LiH. However, highly concentrated acidic solutions are not conducive to easy storage and transportation. This section presents the remarkable results achieved through the thorough mixing of a common solid acid powder (15 wt%) with LiH, followed by the gradual addition of pure water.

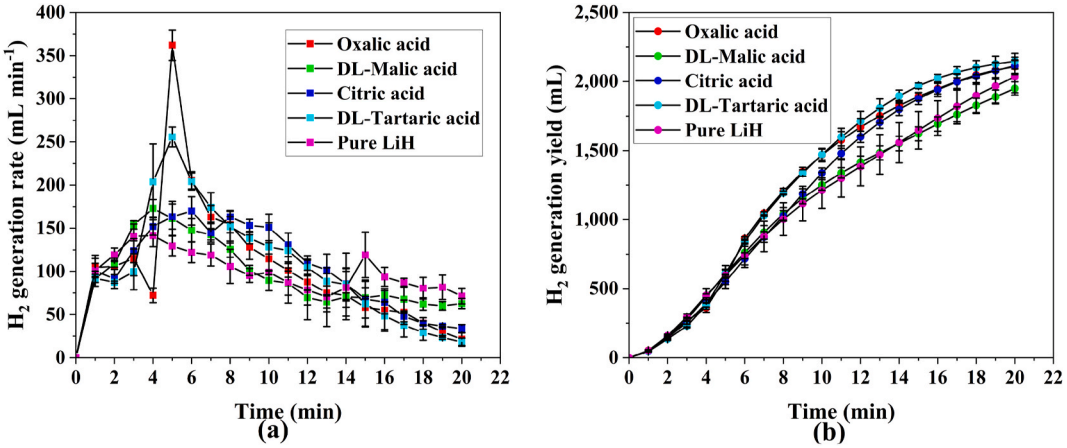


Fig. 3. (a) H_2 generation rate and (b) yield using different solid acids added at 15 wt% relative to LiH at 25 °C with a pump speed of 0.225 mL min^{-1}.

Table 2
PH of 0.15 g solid acid dissolved in 4.5 mL of water.

Acid Type	Temperature (°C)	pH
Oxalic Acid	20	0.7
	30	0.7
	40	0.7
DL-Malic Acid	20	1.8
	30	1.8
	40	1.8
Citric Acid	20	1.7
	30	1.7
	40	1.7
DL-Tartaric Acid	20	1.6
	30	1.6
	40	1.6

Dropping water into the mixture dissolves the solid acid, which ionizes to release hydrogen ions, corroding the resulting LiOH, thereby increasing the contact area between water and LiH to enhance the hydrolysis rate. Table 1 presents the parameters of the various solid acids.

Fig. 3 shows the H_2 generation rates and yields under different solid acid doping conditions. Fig. 3a shows that doping with solid acids resulted in an increase in the hydrolysis rate, reaching a maximum hydrolysis rate of 362.0 mL min^{-1} when doped with oxalic acid, despite significant fluctuations. Conversely, the H_2 generation rates with other acids remained relatively stable. From the H_2 generation rate curves, it can be inferred that the addition of citric acid resulted in minimal fluctuations, with the reaction's maximum rate reaching 170.0 mL min^{-1}. The reaction curves after doping with DL-malic acid and DL-tartaric acids exhibited larger fluctuations than those with citric acid, with maximum H_2 generation rates of 173.1 and 255.8 mL min^{-1}, respectively. Fig. 3b shows that with the addition of 15% solid acid, DL-malic acid doping led to the lowest H_2 generation yield of approximately 1949.4 mL. The H_2 generation yields with DL-tartaric acid, citric acid and oxalic acid increased accordingly to 2147.7, 2114.0 and 2108.3 mL, respectively, larger than that achieved with pure LiH (2035.4 mL). The reason for this situation is that the hydrogen ions from acid ionization corroded LiOH, increasing the contact between water and LiH, thereby increasing the yield.

Table 2 presents the changes in pH when 0.15 g of solid acids were dissolved in 4.5 mL of deionized water at 30, 40, and 50 °C. Data indicates that the pH of the oxalic acid solution was the lowest (0.7), followed by tartaric acid (1.6), citric acid (1.7), and malic acid (1.8). Notably, the pH values remained consistent across various temperature conditions. When the same amount of water was added, the reaction with oxalic acid proceeded faster and exhibited larger fluctuations, which was consistent with the lower pH value of oxalic acid. The pH values of the other three acids were similar and all higher than that measured for oxalic acid, resulting in smaller fluctuations in the rate of

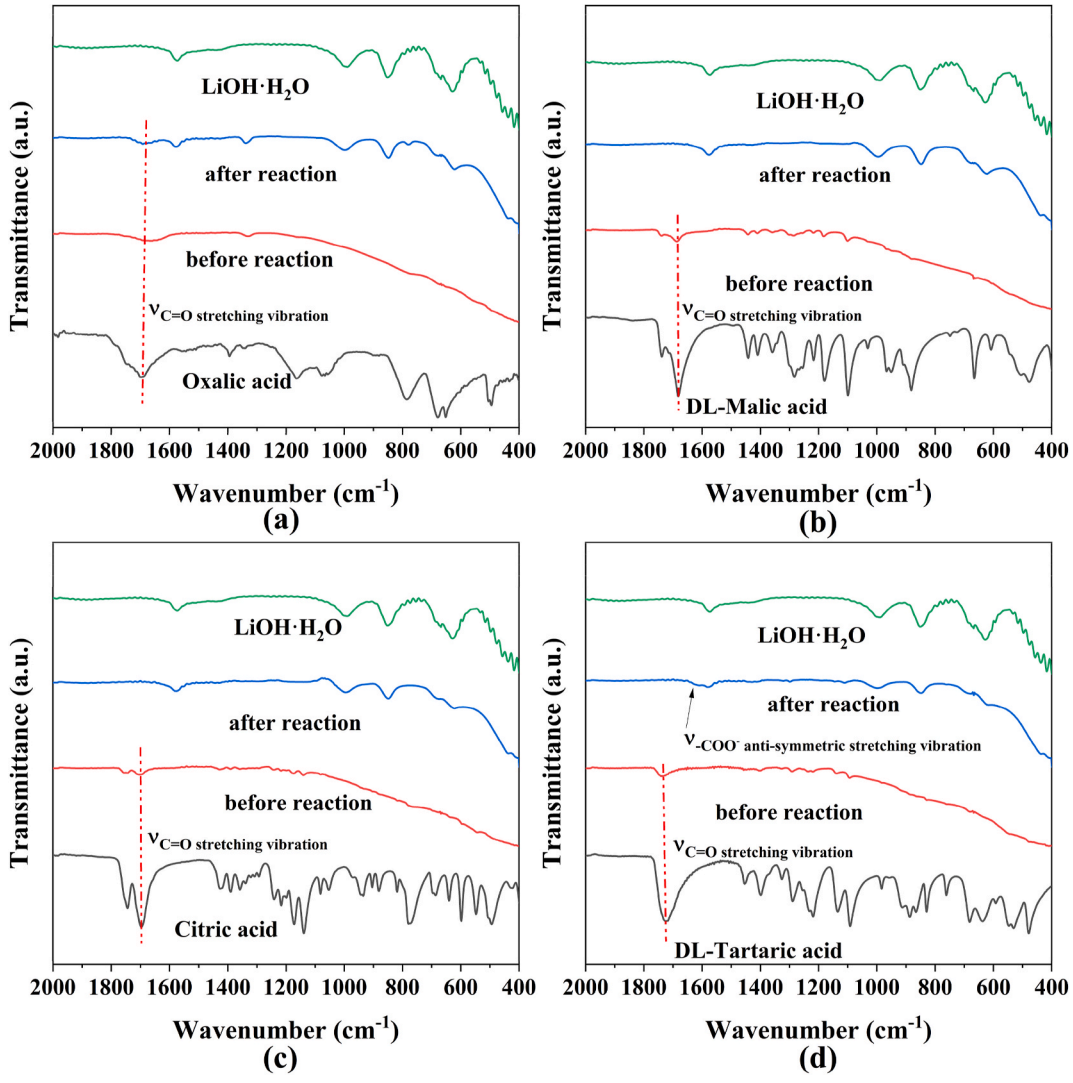


Fig. 4. FT-IR spectra of the mixtures before and after the reaction.

Table 3
Parameters of different strong acid–weak base salts.

Salt Name	Molecular Formula	Molecular Weight (g mol^{-1})	Density (g m^{-3})	Solubility at 20 °C (g/100 g water)
Aluminum chloride	$AlCl_3$	133.34	2.440	45.8
Iron (III) chloride	$FeCl_3$	162.20	2.804	91.8
Copper (II) sulfate	$CuSO_4$	159.61	3.603	20.3

the reaction.

Herein, the XRD results were primarily used to detect inorganic compounds. However, since the solid acids added are organic acids, changes before and after the reaction were monitored using FT-IR spectroscopy. Fig. 4 shows the FT-IR sectra for 15% doping with different solid acids. After thorough mixing, no apparent changes in the position of the characteristic C═O stretching vibration peak of carboxylic acid at about 1700 cm^{-1} were observed compared with that of the pure solid acid, indicating that no reaction occurred during the mixing process. As shown in Fig. 4a, the C═O stretching vibration peak, attributed to the low-solubility oxalic acid, was observed at about 1700 cm^{-1}, corresponding to the hydrolysis reaction product. As shown in Fig. 4d, the anti-symmetric stretching vibration peak of the carboxylate ions ($–COO^-$) observed at about 1610 cm^{-1} was ascribed to the tartaric acid salt product. Only $LiOH·H_2O$ and LiOH were detected in the other two cases. This indicates that under different solid acid doping conditions, all the other acids, except oxalic acid, were completely dissolved.

3.3. Hydrolysis at different strong acid–weak base salts

When salts of strong acid-weak bases dissolve in water, the resulting solution becomes acidic. Strong acid–weak base salts hydrolyze to release hydrogen ions, which then react with LiOH. Theoretically, their effects are similar to those of solid acids. Several of such common salts ($AlCl_3$, $FeCl_3$, and $CuSO_4$) were selected to evaluate their impact on the hydrolysis performance of LiH, at a doping level of 15 wt%. Table 3 presents the parameters of these strong acid–weak base salts.

Fig. 5 shows the H_2 generation rate and yield under different strong acid–weak base salt doping conditions. Fig. 5a shows that the H_2 generation rate is initially accelerated upon the addition of $CuSO_4$, reaching 124.0 mL min^{-1}. Then, the rate decreases before increasing again to a maximum rate of 403.8 mL min^{-1}. By adding $AlCl_3$, the rate curve exhibits a weak peak, with the highest rate reaching 212.5 mL min^{-1}. By adding 15% $FeCl_3$, the maximum rate reaches 221.3 mL min^{-1}. As shown in Fig. 5b, the H_2 generation yields after the addition of $AlCl_3$, $FeCl_3$, and $CuSO_4$, are 2198.9, 2100.0, and 2036.0 mL, respectively. The addition of $AlCl_3$ and $FeCl_3$ yields better results than the H_2 generation yield of pure LiH (2035.4 mL).

Fig. 6 shows the XRD patterns of the mixtures containing 15 wt% of strong acid–weak base salts, before the initiation of the hydrolysis. In the XRD patterns, four characteristic peaks were observed at 38.2°, 44.3°, 64.5°, and 77.5°, corresponding to the (111), (200), (220), and (311) crystal planes of LiH, respectively. For $AlCl_3$, three additional characteristic peaks appeared at 15.1°, 30.4°, and 36.3°, attributed to the (001), (002), and (220) crystal planes of $AlCl_3$, respectively. In the XRD pattern of the $FeCl_3$ mixture, three characteristic peaks at 15.2°, 33.3°, and 43.1° were observed, corresponding to the (003), (11–3), and (11–6) crystal planes of $FeCl_3$. In the case of $CuSO_4$, six characteristic peaks were observed at 21.2°, 25.0°, 34.2°, 37.0°, 39.0°, and 51.3°, attributed to the (101), (210), (220), (301), (221), and (222) crystal planes of $CuSO_4$, respectively. No new phases were formed during the mixing process, indicating that the addition of these salts did not lead to chemical reactions with LiH, and the crystal structures of these salts and LiH remained unchanged under the mixing conditions.

Fig. 7 shows the XRD patterns after the reaction with different strong acid–weak base salts. In the XRD patterns, only five characteristic peaks of $LiOH·H_2O$ were observed, located at 30.1°, 32.0°, 33.7°, 35.0°, and 37.0°, which correspond to its (011), (101), (220), (130), and (– 121) crystal planes, respectively. Additionally, three characteristic peaks at 20.5°, 32.6°, and 35.8° were assigned to the (001), (101), and (110) crystal planes of LiOH, respectively. The solubility of $Al(OH)_3$, $Fe(OH)_3$, and $Cu(OH)_2$, corresponding to the three strong acid–weak base salts, is

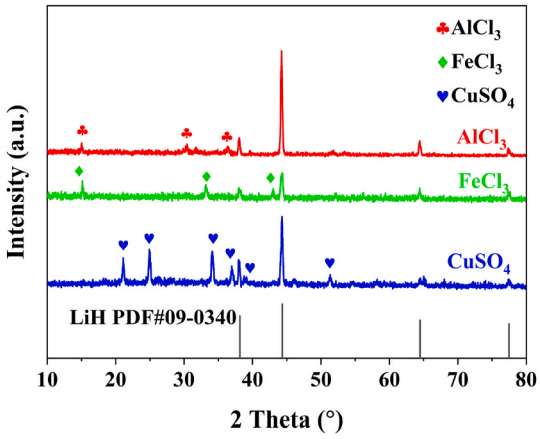


Fig. 6. XRD patterns before the reaction at different strong acid–weak base salts.

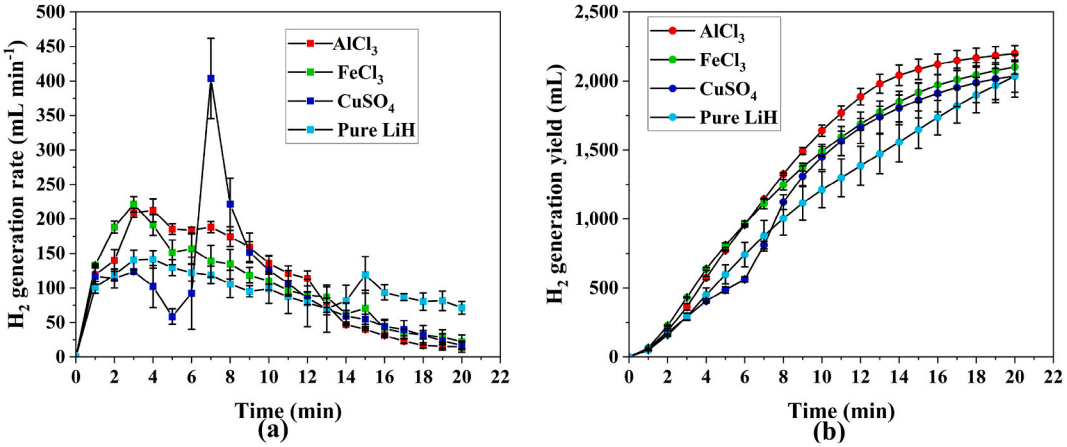


Fig. 5. (a) H_2 generation rate and (b) yield obtained with 15 wt% of different strong acid–weak base salts at 25 °C with a pump speed of 0.225 mL min^{-1}.

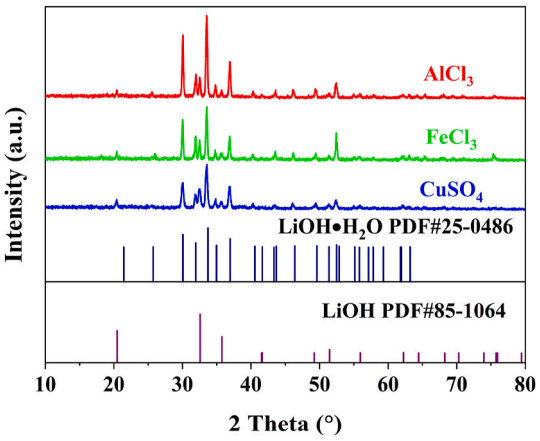


Fig. 7. XRD patterns after the reaction at different strong acid–weak base salts.

significantly lower than that of LiOH. Although these three compounds were expected to be formed during hydrolysis, no compounds containing Al, Fe, and Cu were observed in the XRD patterns. Therefore, XPS was used to analyze the elemental composition of the products.

Fig. 8 presents the XPS spectra after the reaction with different strong acid–weak base salts. The XPS survey spectrum shown in Fig. 8a reveals that the compounds primarily comprised oxygen, carbon, and lithium, with carbon as a contaminant. The high-resolution XPS spectrum of Al 2p shown in Fig. 8b indicated the presence of Al in the hydrolysis product, with a peak at a binding energy of 73.9 eV. Similarly, in the high-resolution XPS spectrum of Fe 2p shown in Fig. 8c, peaks are observed at binding energies of 711.3 and 724.4 eV, corresponding to Fe 2p3/2 and Fe 2p1/2, respectively, indicating the presence of Fe in the hydrolysis product. Moreover, the high-resolution XPS spectrum of Cu 2p shown in Fig. 8d exhibits a peak at the binding energy of 935.6 eV, corresponding to Cu 2p3/2, confirming the presence of Cu in the hydrolysis product.

The presence of elements was determined by XPS; however, the elemental content was low, and LiOH interfered with the detection of the added salt product. To prevent LiOH interference, the following treatments were performed on the product: First, the post-reaction product was washed multiple times with anhydrous methanol and filtered. Next, the filtered material was washed in deionized water. Finally, the insoluble residue was dried via boiling evaporation to eliminate water. Fig. 9 shows the XRD patterns and Raman spectrum of the insoluble residue after the hydrolysis products underwent additional treatments.

As shown in Fig. 9a, three characteristic peaks were observed at 11.7°, 23.5°, and 35.4°, corresponding to the (003), (006), and (111) crystal planes of $LiAl_2(OH)_7{\cdot}2H_2O$. The observations suggest that the following reactions occurred during hydrolysis:

$$Al^{3+} + 3LiOH \rightarrow Al(OH)_3 + 3Li^+$$

$$LiOH + 2Al(OH)_3 \rightarrow LiAl_2(OH)_7$$

As shown in Fig. 9b, four characteristic peaks were observed at 33.3°, 35.7°, 49.5°, and 54.2°, corresponding to the (104), (110), (024), and (116) crystal planes of Fe_2O_3. This indicated the presence of Fe_2O_3 in the insoluble matter, despite the low peak intensities in the XRD pattern. The XRD pattern is consistent with the presence of an amorphous material, suggesting the potential presence of $Fe(OH)_3$, which is known to be amorphous.

The Raman spectrum of the hydrolysis product treated with $FeCl_3$ is

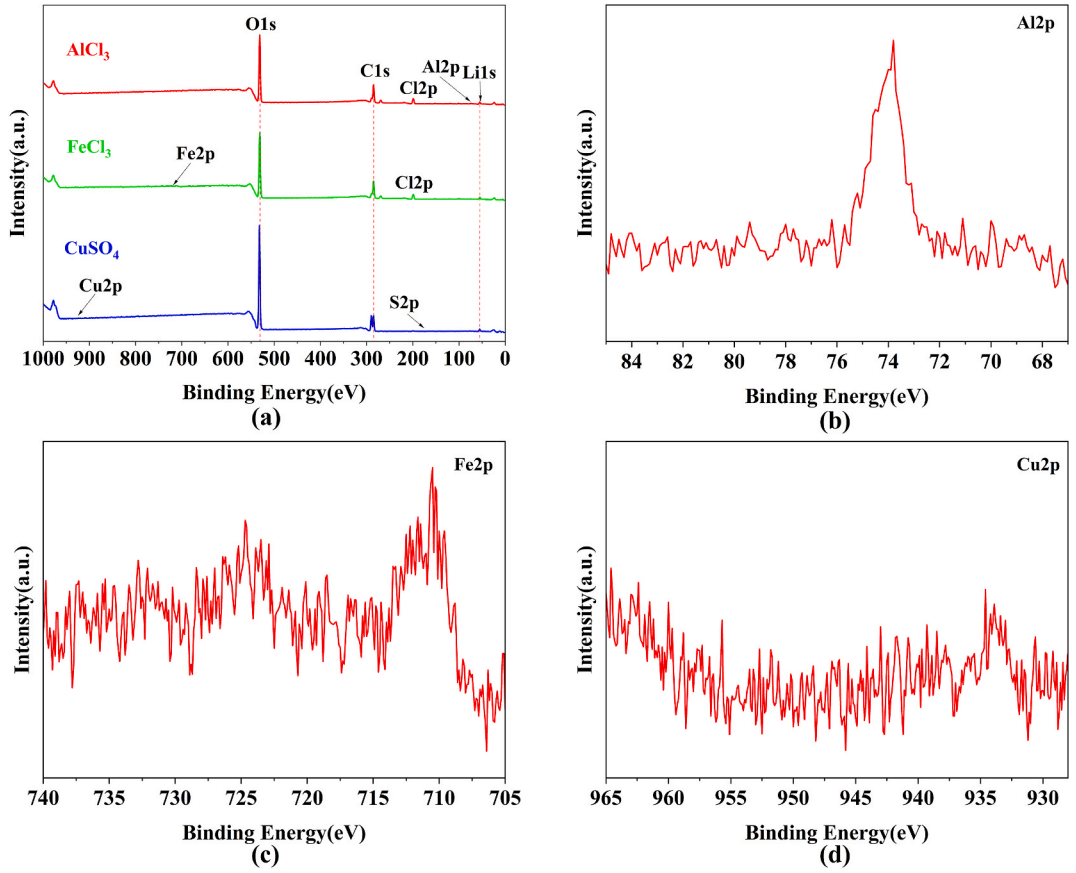


Fig. 8. (a) XPS survey spectrum of the product at different strong acid–weak base salts, (b) Al 2p XPS spectrum of the hydrolysis product with $AlCl_3$, (c) Fe 2p XPS spectrum of the hydrolysis product with $FeCl_3$, and (d) Cu 2p XPS spectrum of the hydrolysis product with $CuSO_4$.

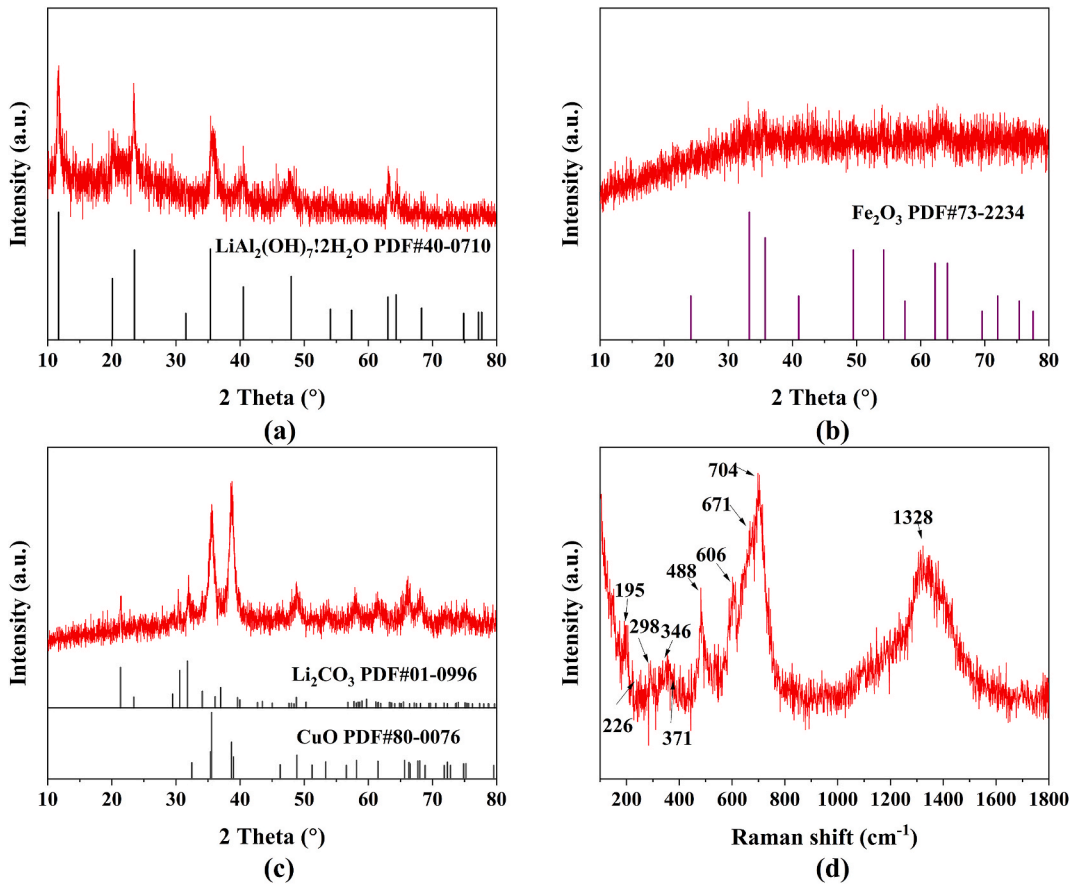


Fig. 9. XRD patterns of the hydrolysis products after treatment with different salts: (a) $AlCl_3$, (b) $FeCl_3$, and (c) $CuSO_4$. (d) Raman spectrum after treatment of the hydrolysis product with $FeCl_3$.

shown in Fig. 9d. The spectrum contained seven peaks at 195, 298, 346, 488, 606, 704, and 1328 cm^{-1}, which can be attributed to iron oxide, and peaks at 226, 371, and 671 cm^{-1}, which can be attributed to iron hydroxide. Compared with the XRD pattern, the Raman spectrum exhibited stronger signals for Fe_2O_3 than for $Fe(OH)_3$. This can be attributed to the decomposition of $Fe(OH)_3$ under the high temperature of the laser during the Raman measurements. The presence of Fe_2O_3 and $Fe(OH)_3$ indicates the following reaction during the hydrolysis process: $Fe(OH)_3$ precipitates from trivalent iron ions and subsequently dehydrates to form Fe_2O_3. The following reactions occur during the hydrolysis process:

$$Fe^{3+} + 3LiOH \rightarrow Fe(OH)_3 + 3Li^+$$

$$2Fe(OH)_3 \xrightarrow{\Delta} Fe_2O_3 + 3H_2O$$

As shown in Fig. 9c, four characteristic peaks appeared at 35.4°, 35.5°, 38.6°, and 39.0°, corresponding to the (002), (−111), (111), and (200) crystal planes of $CuSO_4$. Additionally, peaks at 21.3°, 30.6°, and 31.8° were attributed to the (110), (110), and (002) crystal planes of Li_2CO_3. During hydrolysis, copper ions precipitate as $Cu(OH)_2$, which is then converted to CuO upon heating. The basic reactions are as follows: Li_2CO_3 forms through the facilitated solution adsorption of CO_2 by CuO [45].

$$Cu^{2+} + 2LiOH \rightarrow Cu(OH)_2 + 2Li^+$$

$$Cu(OH)_2 \xrightarrow{\Delta} CuO + H_2O$$

4. Conclusions

This study investigated methods to mitigate the impact of LiOH formation on the hydrogen production performance in liquid water hydrolysis of LiH. The LiOH generated by the hydrolysis of LiH encapsulates the unreacted LiH, hindering the contact between water and LiH, thus reducing the hydrolysis rate of LiH. Dilute sulfuric acid solutions with hydrogen ion concentrations of c $(H^+) = 1$, 0.1, and 10^{-4} mol L^{-1} were used to hydrolyze LiH. As the concentration of hydrogen ions increased, the corrosion and dissolution of LiOH increased, accelerating LiH hydrolysis. The hydrogen ion concentration is 1 mol L^{-1}, and the fastest hydrolysis rate that can be achieved is 232.6 mL min^{-1}. The solid acid doped in LiH dissolves in water and ionizes, releasing hydrogen ions. The hydrogen ions corrode LiOH, achieving a similar effect to an acidic solution. The hydrolysis rate of LiH increased as the acidity of the acid solution formed upon dissolving a unit mass of acid in water increased. Doping with oxalic acid achieved the fastest hydrogen generation rate of 362.0 mL min^{-1}, with a hydrogen generation yield of 2108.3 mL. The strong acid -weak base salt doped in lithium hydride, when dissolved in water, undergoes hydrolysis and produces hydrogen ions. The alkaline environment created by the hydrolysis of LiH enhanced the hydrolysis process of strong acid -weak base salt, and hydrogen ions reacted with the LiOH product, thereby increasing the rate of LiH hydrolysis. Doping with $CuSO_4$ achieved the fastest hydrogen generation rate of 403.8 mL min^{-1}, with a hydrogen generation yield of 2036.0 mL. The experimental results show that an increase in the concentration of hydrogen ions and the addition of solid acids and strong acid–weak base salts are effective strategies for reducing the adverse effects of LiOH formation during LiH hydrolysis for hydrogen production. In our upcoming research, our primary emphasis will be on regulating of rate of LiH hydrolysis and investigating its performance under various solid acid and strong acid–weak base salt conditions.

CRediT authorship contribution statement

Jiaxu Zhou: Conceptualization, Investigation, Validation, Writing – original draft, Writing – review & editing, Data curation. **Huichao Deng:** Investigation, Methodology, Validation, Writing – original draft. **Liwei Yang:** Conceptualization, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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